

Service-Oriented Computing Kit

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Abstract

This paper describes a Service-Oriented Computing Kit (SOCK), which is an overarching framework covering the key artifacts in planning, modeling, designing, developing, deploying, and managing service-oriented solutions in the enterprise computing space. Based on a divide-and-conquer strategy, this comprehensive kit is a systematic taxonomy to abstract complexities and organize the major aspects of service-oriented development, so that the roles, responsibilities, accountabilities, skillsets, procedures and deliverables can be clearly defined for the IT teams to effectively construct service-oriented systems. *SOCK* comprises eight modules – *Architecture*, *Process*, *Integration*, *Environment*, *Technology*, *Development*, *Management*, and *Roadmap*. Each module addresses specific technical concerns in particular areas. *SOA* provides the service definitions, service catalog, service composition, scope of applications and portfolios, and service architecture modeling methodology. *SOP* deals with the business process management, process modeling & notations, process orchestration, coordination/collaborations, human interactions, and 1-phase commit compensation for long-lived processes. *SOI* concentrates on the service interoperability, integration patterns, enterprise service bus, traditional synchronous/asynchronous enterprise application integration, and integration with portals and content management systems. *SOE* consists of the runtime infrastructure, service registry for discovery and directory, service transformation/routing gateway, service virtualization via grid computing, and quality-of-services compliance with service-level agreements. *SOT* covers the implementation technologies – web services, standards and specifications, technical patterns, convergence, and aspect-oriented techniques. *SOD* is composed of the development lifecycle, programming model, design/development tools, frameworks, reusable utilities/toolkits/components, and model-driven approaches. *SOM* includes services & policies repository in service identification and categorization, security mechanisms, business activity monitoring, reporting, BI, and provisioning. *SOR* handles the planning/blueprint, strategy, KPI dashboards, governance, and organization. The elements in each module are further decomposed to form loose semantic associations. This holistic kit has been implemented on an open source mind-mapping platform, which is used in guiding the migration from the conventional multi-tier web application style to a service-oriented paradigm in the financial services sector, and *SOCK* has proven to be an effective approach to pragmatically transitioning to the new computing mode.

Keywords

Service-oriented, computing, kit, framework, model, architecture, process, integration, environment, technology, development, management, roadmap, infrastructure, standards, practice, pattern, grid, taxonomy.

About the authors

Tony Shan is a Chief Architect/Strategist at Wachovia Bank and adjunct professor at University of Phoenix. He has been working on computing technologies for 20+ years, with extensive experience in a variety of technologies and programming languages in different industries. He has initiated/directed advanced research on emerging technologies, resulting in an invention patent and award-winning IT solutions as well as several methodologies and platform models for adaptive system development. He has led establishing IT strategies and architecture blueprints, coupled with pragmatic technology roadmaps and enterprise architecture standards/policies, for Fortune 100 international organizations. In addition to dozens of top-notch technical publications, he authored multiple books on asynchronous web services, heterogeneous business integration, application frameworks and data caching.

Winnie Hua is a Principal Consultant in CTS Inc. She has more than 15 years project and consulting experience on a broad range of leading-edge technologies. As a solutions architect/lead, she has directed lifecycle design and development of large-scale eCommerce systems on diverse platforms using a variety of cutting-edge technologies and unified/agile methodologies. She has also initiated advanced research on various emerging web technologies. She is a regular speaker in conferences, and also a co-founder of Charlotte Architecture & Technology Symposium.